

NUTRITION IN BACTERIA

Fully Classified & Identifiable Study Guide - Types - Differences - Diagrams - Similarity / Difference / Synonym - 4-Mark & 4.8-Mark Blocks

Every bacterium is described by **WHOSE energy it uses + WHICH carbon it fixes + WHICH electron it oxidizes**. Master this triad and every type becomes identifiable.

How to read this guide

Bacterial nutrition is answered by **3 questions**. A bacterium's "nutritional type" is just a code word for its answers:

Question	Possible answer 1	Possible answer 2	Root word
1. Energy source (where does it get power?)	Light	Chemicals (reduce a donor)	Photo- / Chemo-
2. Carbon source (what does it build cells from?)	CO ₂ (inorganic)	Organic (glucose etc.)	-autotroph / -heterotroph
3. Electron / energy donor (what gets oxidized?)	Inorganic (H ₂ , NH ₃ , H ₂ S, Fe ²⁺)	Organic (sugars, fatty acids)	-lithotroph / -organotroph

Mnemonic "E-C-E": Energy - Carbon - Electron-donor = the 3 letters that name every bacterium. e.g. Cyanobacteria = **P(hoto)-A(uto)-L(itho)troph**; a sugar bug = **C(chemo)-H(etero)-O(rgano)troph**.

The MASTER Classification Tree (draw this in exam)

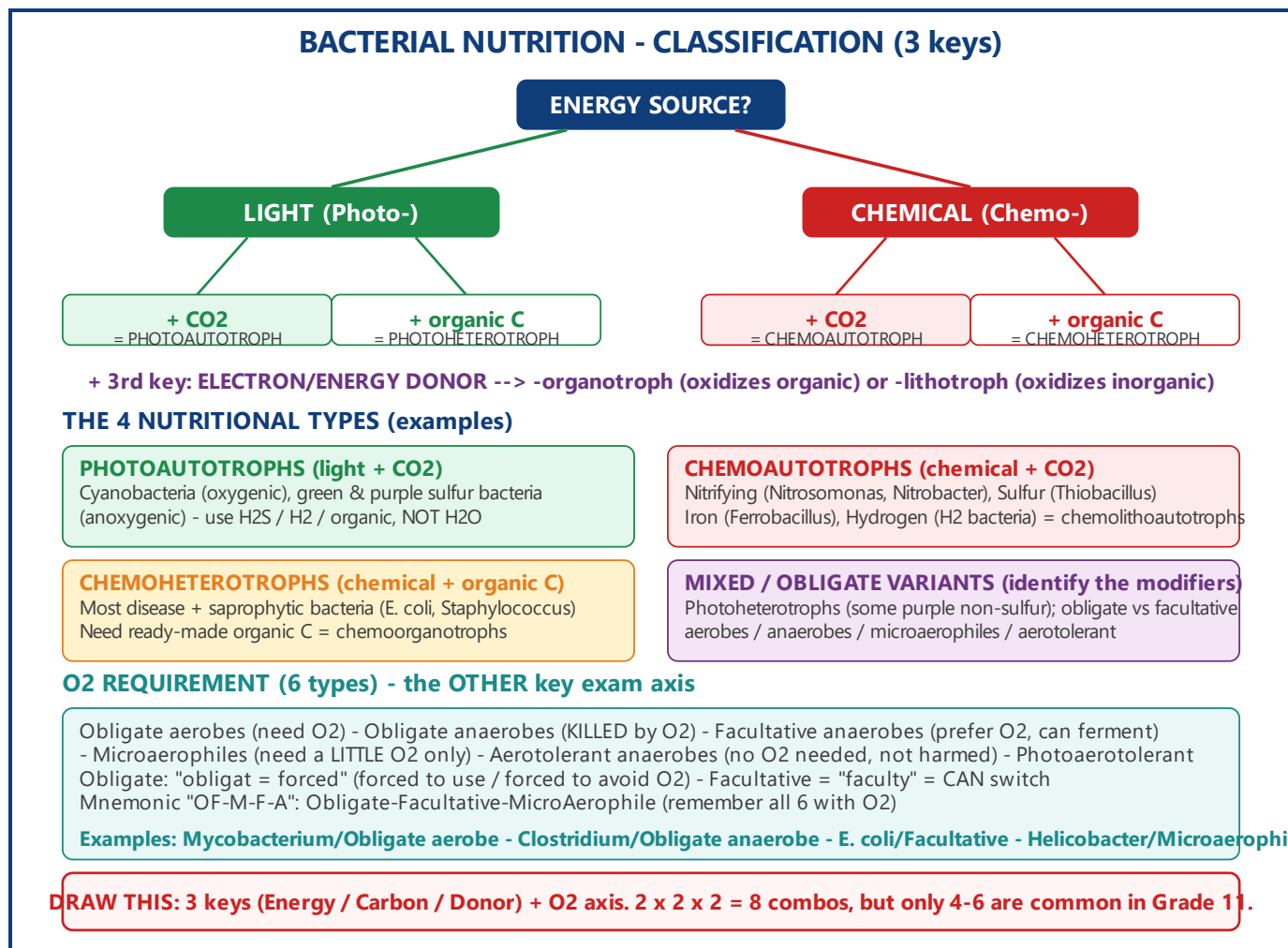


Fig 1: The master classification tree - energy x carbon x donor, plus the 6 O₂-requirement types. This one figure covers the whole chapter.

The 4 Basic Nutritional Types - details + differences

What this means (plain): A bacterium is *auto* if it makes its own food from CO₂ (like a plant) and *hetero* if it must eat ready organic food (like you). *Photo* = uses sunlight; *chemo* = uses chemicals. So: plants & cyanobacteria = photo-autotrophs; most bugs & disease bacteria = chemo-heterotrophs; soil "cleaners" (nitrifiers) = chemo-autotrophs.

Type	Energy	Carbon	Electron donor	Examples
Photoautotroph	Light	CO ₂	H ₂ O (cyanob.) / H ₂ S (sulfur)	Cyanobacteria, green & purple S bacteria
Chemoautotroph (chemolithoautotroph)	Chemical (inorganic)	CO ₂	NH ₃ , H ₂ S, Fe ²⁺ , H ₂	Nitrosomonas, Nitrobacter, Thiobacillus
Photoheterotroph	Light	Organic	Organic	Purple non-sulfur bacteria

Chemoorganotroph (chemoheterotroph)	Chemical (organic)	Organic	Organic	E. coli, Staphylococcus, Lactobacillus
--	-----------------------	---------	---------	---

Similarity / Difference / Synonym: All photo-types use a pigment (Chl a, bacteriochlorophyll, carotenoid).

Difference: cyanobacteria split H_2O → release O_2 (oxygenic); sulfur bacteria split H_2S → deposit S^0 (anoxygenic, no O_2). SYNONYMS: chemolithoautotroph = chemoautotroph = inorganic-food maker; chemoorganotroph = chemoheterotroph = organic-food eater.

PROBABLE 4-MARK - "Differentiate 4 nutritional types of bacteria" 4

- Photoautotroph: light + CO_2 ; e.g. Cyanobacteria, green/purple sulfur bacteria = 1.
- Chemoautotroph (chemolithoautotroph): oxidizes inorganic $NH_3/H_2S/Fe$ for energy + CO_2 ; e.g. Nitrifiers = 1.
- Photoheterotroph: light + organic carbon; purple non-sulfur = 1.
- Chemoorganotroph (chemoheterotroph): organic for energy + carbon; most pathogenic/saprophytic = 1.

4.8-MARK EXTENSION

+ state each type's ELECTRON DONOR (H_2O / H_2S / inorganic / organic) and one ecological role (fixing C / nitrification / sulfur cycle / decomposition). Draw the classification tree (Fig 1) with the 4 types labelled (+0.8).

A - PHOTOAUTOTROPHS (light + CO_2)

What this means (plain): These bacteria are "plant-like" - they trap sunlight with pigments and fix CO_2 into food (photoautotrophy). Two big groups: **oxygenic** (cyanobacteria - split water, release O_2) and **anoxygenic** (green & purple sulfur bacteria - split H_2S , trap S as granules, NO O_2). This is THE exam distinction.

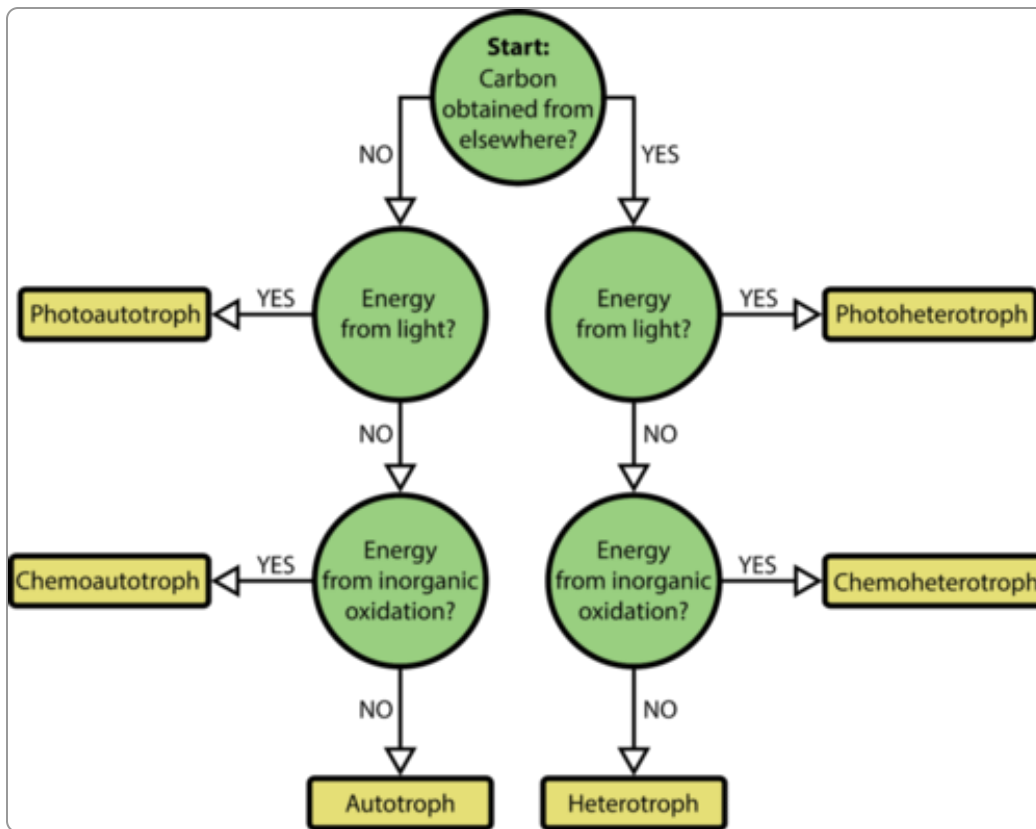


Fig 2: Bacterial nutrition overview (photo- vs chemo- autotrophs & heterotrophs)

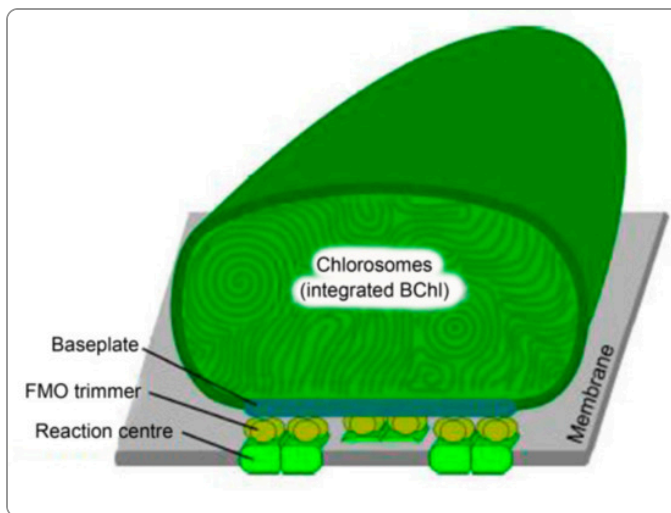


Fig 3: Anoxygenic phototrophs - green & purple sulfur bacteria pigments

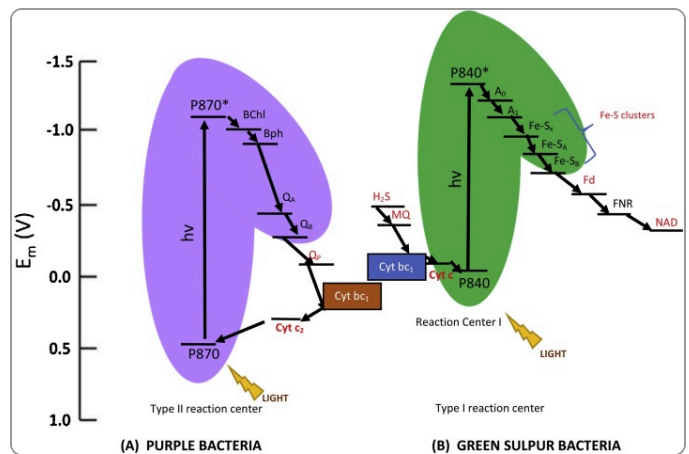


Fig 4: Anoxygenic phototrophic bacteria (purple non-sulfur, green non-sulfur)

Group	Oxygenic? (splits H ₂ O)	Pigment	Donor	Examples
Cyanobacteria (blue-green algae)	YES - releases O ₂	Chl a + phycobilin	H ₂ O	Nostoc, Anabaena, Oscillatoria, Spirulina
Green sulfur (Chlorobiaceae)	NO - H ₂ S / S ²⁻	Bacteriochlorophyll a + carotene	H ₂ S, S ²⁻ , S	Chlorobium, Chlorothrix

Purple sulfur (Chromatiaceae)	NO - H ₂ S	Bacteriochlorophyll a/g	H ₂ S	Chromatium, Thioploca (S granules)
Purple non-sulfur (Rhodospirillaceae)	NO - organic donor (photo-HETERO)	Bacteriochlorophyll	Organic	Rhodospirillum, Rhodocystis
Green non-sulfur (Chloroflexaceae)	NO - organic donor (photo-HETERO)	BChl + Chl b	Organic	Chloroflexus

Similarity / Difference / Synonym: ALL use a light-harvesting pigment + electron transport (redox). Difference: *oxygenic* vs *anoxygenic* = H₂O (O₂) vs H₂S (no O₂, S deposits). SYNONYMS: photoautotroph = photolithoautotroph (when H donor is inorganic); "blue-green algae" = cyanobacteria (bacteria, NOT algae).

PROBABLE 4-MARK - "Oxygenic vs Anoxygenic photoautotrophs (4 points)" 4

- Oxygenic: splits H₂O, releases O₂; Chl a; e.g. Cyanobacteria = 1.
- Anoxygenic: splits H₂S/S, NO O₂; bacteriochlorophyll; e.g. green/purple S = 1.
- Oxygenic thylakoids = in granum; anoxygenic = membrane infoldings / chlorosomes = 1.
- Cyanobacteria have heterocysts (fix N₂); purple S store S granules = 1.

PROBABLE 4-MARK - "Name 4 photoautotrophic bacteria & state their O₂ type" 4

- Nostoc / Anabaena = cyanobacteria, OXYGENIC = 1.
- Chlorobium (green S) = anoxygenic = 1.
- Chromatium (purple S) = anoxygenic, S granules = 1.
- Rhodospirillum (purple non-S) = photoHETEROTROPH, anoxygenic = 1.

B - CHEMOAUTOTROPHS / CHEMOLITHOTROPHS (oxidize inorganic for energy + CO₂)

What this means (plain): These bacteria get their energy by **oxidizing inorganic minerals** (ammonia, hydrogen sulfide, ferrous iron, hydrogen gas) and build cells from CO₂. They are the "soil cleaners" - **nitrifiers** (turn NH₃ -> nitrate) and **sulfur bacteria** (turn H₂S -> sulfate). They drive the N and S cycles.

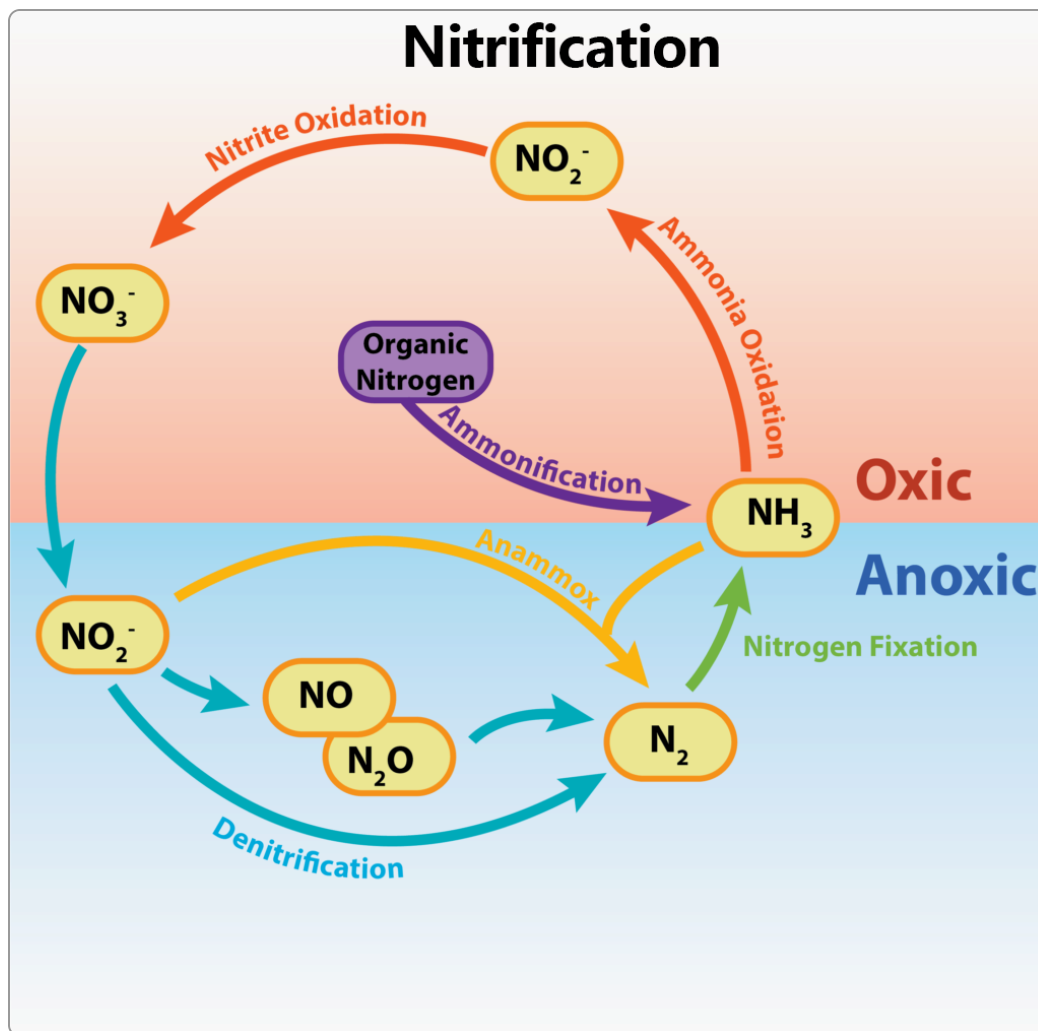


Fig 5: Chemolithotrophy & nitrogen metabolism (nitrifying & S-oxidizing pathways)

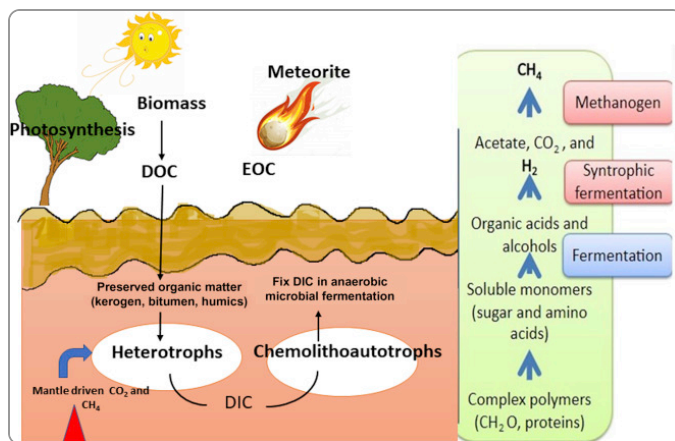


Fig 6: Chemolithoautotrophs (energy from inorganic oxidation)

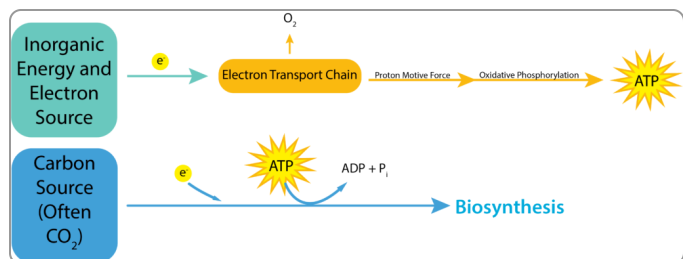


Fig 7: Chemolithotrophy & N metabolism (LibreTexts)

Type	Inorganic oxidized	Product	Examples
Nitrogen (nitrifying)	$\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$	Nitrite then nitrate	Nitrosomonas, Nitrobacter
Sulfur	$\text{H}_2\text{S} / \text{S} / \text{S}^{2-}$	Sulfate (SO_4^{2-})	Thiobacillus, Beggiatoa, Thiothrix

Iron	Fe^{2+}	Fe^{3+}	Ferrobacillus (Ferrobacterium)
Hydrogen	H_2	H_2O	H_2 bacteria, Rickettsia
Methanogenic (archaea)	$\text{H}_2 + \text{CO}_2$ / acetate	CH_4	Methanogens (strict anaerobes)

Similarity / Difference / Synonym: All chemoautotrophs: inorganic energy + CO_2 carbon + aerobic (need O_2) except H_2 & methanogens (anaerobic). Difference: N-oxidizers make nitrate; S-oxidizers make sulfate; Fe-oxidizers make Fe^{3+} . **SYNONYMS:** chemolithoautotroph = chemoautotroph = lithoautotroph; "nitrification" = $\text{NH}_3 \rightarrow \text{NO}_2^-$ (Nitrosomonas) + $\text{NO}_2^- \rightarrow \text{NO}_3^-$ (Nitrobacter).

PROBABLE 4-MARK - "Nitrification: the 2 steps with organisms" 4

- Step 1: $\text{NH}_3 \rightarrow \text{NH}_4^+ \rightarrow \text{NO}_2^-$ (nitrite) by Nitrosomonas / Nitrosococcus = 1.
- Step 2: $\text{NO}_2^- \rightarrow \text{NO}_3^-$ (nitrate) by Nitrobacter / Nitrospira = 1.
- Both are aerobic CHEMOLITHOAUTOTROPHS (CO_2 + inorganic NH_3) = 1.
- Significance: makes N plant-available; key in the N cycle & soil = 1.

4.8-MARK EXTENSION

+ a nitrifying (NH_3 oxidation) reaction and one sulfur-oxidation reaction; state the ecological role (N/S cycle, bioremediation). Draw Fig 5 and label the 2 nitrifying + 1 sulfur bacterium (+0.8).

C - CHEMOHETEROTROPHS / CHEMOORGANOTROPHS (organic energy + organic carbon)

What this means (plain): The MOST common type. They must eat ready organic food (glucose, amino acids) for both energy AND carbon. This group includes **saprophytes** (decomposers), **parasites** (live in/on a host) and **symbionts** (mutual). The exam often asks you to split chemo-heterotrophs into these 3 lifestyles.

Lifestyle	Carbon / energy source	Examples
Saprophyte (decomposer / mineralizer)	Dead organic matter	Most soil bacteria, Lactobacillus (fermentation)
Parasite / pathogen	Living host tissue	Staphylococcus, Salmonella, Mycobacterium
Symbiont (mutualistic)	Host, in return for service	Rhizobium (N_2 fixation in root nodules)

Metabolic fates of glucose (3 ways a chemoorganotroph makes ATP)

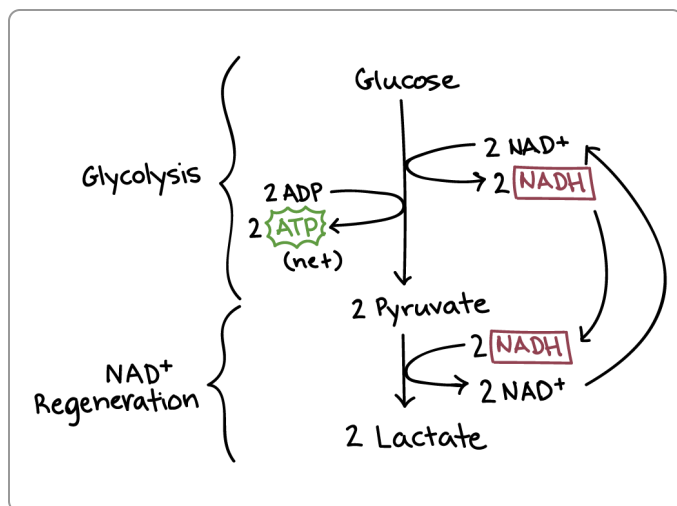


Fig 8: Aerobic respiration vs fermentation vs anaerobic respiration

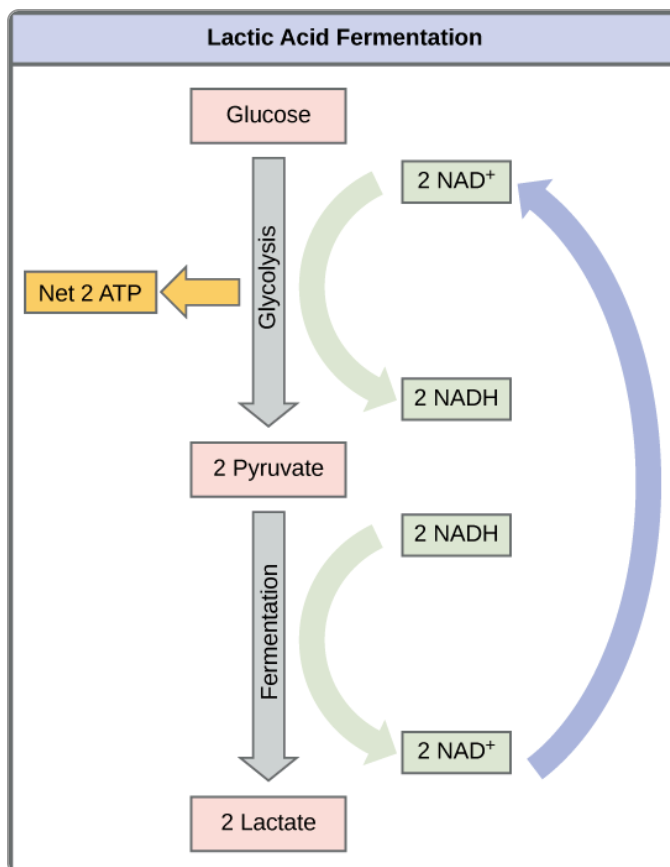


Fig 9: Fermentation (lactic / ethanol) energy yield

Process	Electron acceptor	Yield	Products	Example
Aerobic respiration	O ₂	High (~30-38 ATP/glucose)	CO ₂ + H ₂ O	Most E. coli, Bacillus
Anaerobic respiration	NO ₃ ⁻ /SO ₄ ²⁻ /Fe	Moderate	NO ₂ ⁻ /H ₂ S/FeS	Denitrifiers (Clostridium)
Fermentation	Organic (e.g. pyruvate)	Low (2 ATP)	Lactic / ethanol + CO ₂	Lactobacillus, yeast

Similarity / Difference / Synonym: All chemoorganotrophs: organic = energy + carbon; differ by O₂ need and electron-acceptor. Difference: respiration (O₂ or NO₃⁻) vs fermentation (organic acceptor, no ETC).
SYNONYMS: chemoorganotroph = chemoheterotroph = organotroph; "decomposer = mineralizer = saprophyte"; denitrification = NO₃⁻ → N₂ (Clostridium, Pseudomonas).

PROBABLE 4-MARK - "3 ways bacteria convert glucose (respiration types)" 4

- Aerobic respiration: O₂ acceptor, CO₂+H₂O, high ATP = 1.
- Anaerobic respiration: NO₃⁻/SO₄²⁻ acceptor, partial CO₂, moderate ATP = 1.
- Fermentation: organic acceptor, lactic/ethanol + CO₂, 2 ATP = 1.
- State one organism per type (E. coli / denitrifiers / Lactobacillus) = 1.

4.8-MARK EXTENSION

+ a word equation for aerobic respiration ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$) and one lactic-fermentation equation; state ATP yields. Draw Fig 8 (+0.8).

PROBABLE 4-MARK - "Saprophytic vs parasitic vs symbiotic (distinguish)" 4

- Saprophyte: feeds on DEAD organic matter = decomposer (e.g. soil bacteria, Lactobacillus) = 1.
- Parasite: feeds on a LIVING host; causes disease (e.g. Staphylococcus, Mycobacterium) = 1.
- Symbiont: mutualistic - both benefit (e.g. Rhizobium fixes N_2 for the legume) = 1.
- All three are CHEMOHETEROTROPHS (organic C) but differ in host relationship = 1.

D - O_2 REQUIREMENT TYPES (the 2nd exam axis) + SPECIAL METABOLISMS

What this means (plain): Apart from WHAT it eats, every bacterium is also classed by how it deals with O_2 . "Obligate aerobe" = must have O_2 ; "obligate anaerobe" = O_2 kills it; "facultative anaerobe" = prefers O_2 but can switch to fermentation; "microaerophile" = needs a little O_2 ; "aerotolerant" = doesn't need O_2 but isn't harmed.

Type	Uses O_2 ?	Harmed by O_2 ?	Examples
Obligate aerobe	YES (needs it)	No	Mycobacterium, Pseudomonas
Obligate anaerobe	No	YES (kills it)	Clostridium, Methanogens
Facultative anaerobe	Yes (prefers), but can ferment	No	E. coli, Staphylococcus
Microaerophile	A LITTLE O_2 only	Killed by high O_2	Helicobacter, Campylobacter
Aerotolerant	No (fermentative)	No (tolerates)	Lactobacillus, Streptococcus
Photoaerotolerant	No (phototroph)	No	Green & purple non-sulfur

Mnemonic "O-OF-M-A" (the 5 O_2 types): **O**bligate-aerobe, **O**bligate-anaerobe, **F**acultative, **M**icroaerophile, **A**erotolerant. "O-OF-M-A" = *O*bligates, *F*acultative, *M*icro, *A*ero. Obligate = forced; facultative = can switch; micro = small dose; aero-tolerant = not harmed.

Special N-metabolisms (frequently tested with nutrition)

Process	Reaction	Organisms
N_2 fixation	$N_2 \rightarrow NH_3$ (via nitrogenase)	Rhizobium, Azotobacter, Clostridium, cyanobacteria (Nostoc)

Nitrification	$\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$	Nitrosomonas, Nitrobacter
Denitrification	$\text{NO}_3^- \rightarrow \text{N}_2 / \text{N}_2\text{O}$	Clostridium, Pseudomonas, Thiobacillus

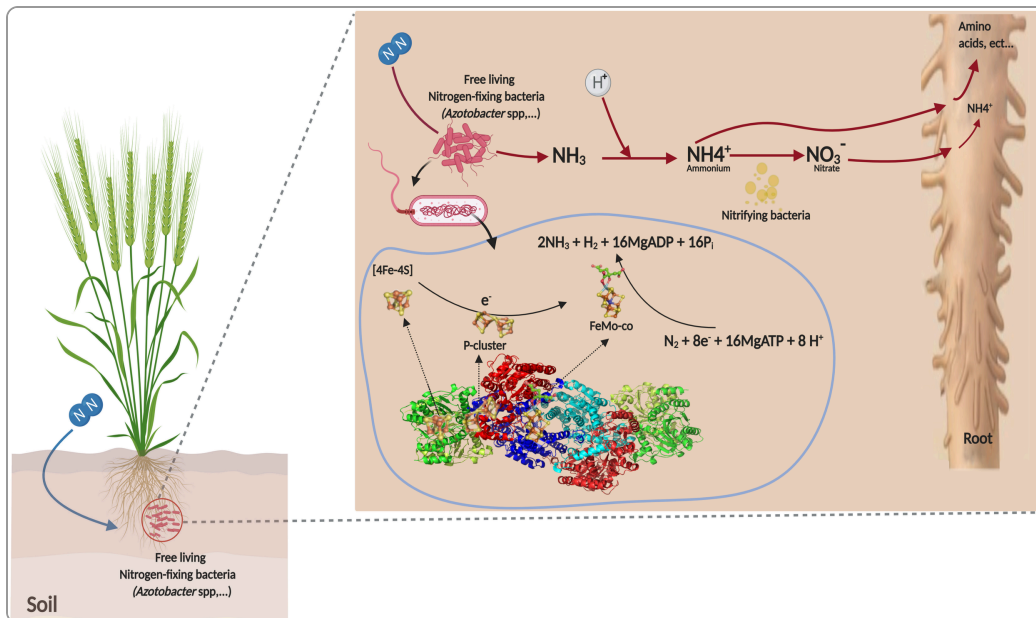


Fig 10: N_2 -fixing *Azotobacter* (soil) - symbiotic + free-living N fixation

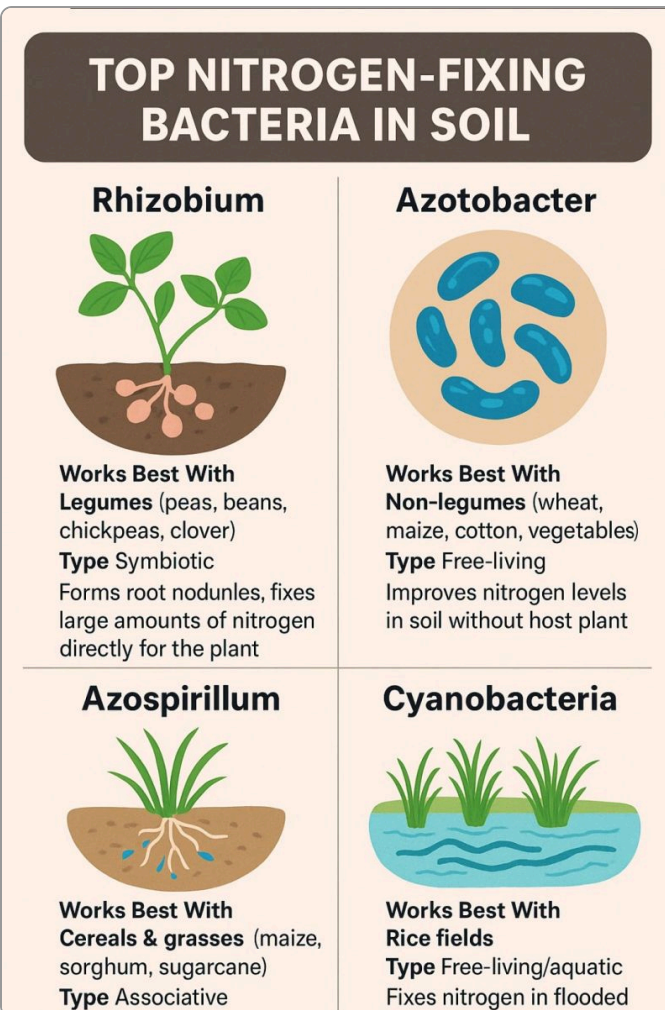


Fig 11: Soil N_2 -fixing bacteria compared

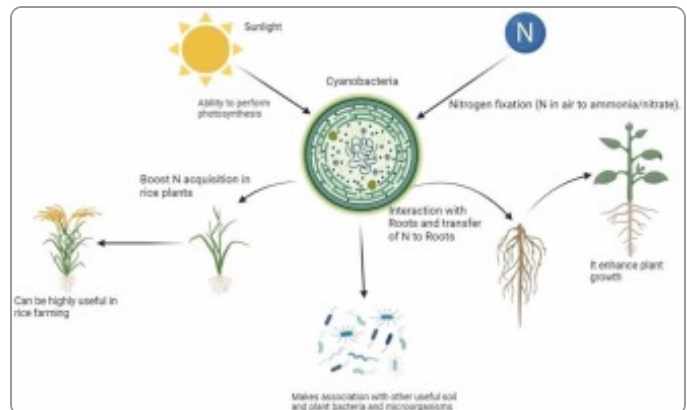


Fig 12: N_2 -fixing cyanobacteria (*Nostoc* / *Anabaena*)

Similarity / Difference / Synonym: N_2 fixation = $N_2 \rightarrow NH_3$ (adds N); nitrification = $NH_3 \rightarrow NO_3^-$ (cleans N); denitrification = $NO_3^- \rightarrow N_2$ (loses N to air). NITROGENASE works ONLY anaerobically (blocked by O_2) $\rightarrow$ cyanobacteria fix N_2 in **heterocysts**. SYNONYMS: symbiont = mutualistic N-fixer; "nitrogenase = N_2 -ase" = the O_2 -sensitive enzyme.

PROBABLE 4-MARK - " N_2 fixation: organisms + conditions" 4

- Reaction $N_2 + 8H + 2e^- + 16ATP \rightarrow 2NH_3 + 14ATP + 2ADP$ (needs ATP + NITROGENASE) = 1.
- Symbiotic: Rhizobium (legume nodules); free-living: Azotobacter, Clostridium; cyanobacteria (Nostoc) = 1.
- Nitrogenase is O_2 -SENSITIVE $\rightarrow$ cyanobacteria fix N_2 in HETEROCYSTS (no O_2) = 1.
- Significance: adds N to soil = natural fertilizer = 1.

4.8-MARK EXTENSION

+ heterocyst structure (thick wall, no PSII, O_2 -free) + one free-living vs one symbiotic N-fixer with a habitat. Draw Fig 12 + label heterocyst + a vegetative cell (+0.8).

PROBABLE 4-MARK - "6 O_2 types (name & example each)" 4

- Obligate aerobe (Mycobacterium) + Obligate anaerobe (Clostridium) = 1.
- Facultative anaerobe (E. coli) = 1.
- Microaerophile (Helicobacter) = 1.
- Aerotolerant (Lactobacillus) - 4 types named + role = 1.

ONE-PAGE QUICK-IDENTIFY (all types as short outlets)

BACTERIAL NUTRITION - QUICK IDENTIFY KEY (all types)

A. PHOTOAUTOTROPHS (light + CO₂)

- Oxygenic: cyanobacteria (Nostoc, Anabaena) = O₂ released
- Anoxygenic green S (Chlorobium), purple S (Chromatium) = H₂S, S granules
- Purple/GREEN non-S (Rhodospirillum/Chloroflexus) = photo-HETERO (light + organic C)
- Pigments: Chl a + phycobillin (oxygenic) / BChl (anoxygenic)
- ID: "plants" = oxygenic; "S granules + purple" = anoxygenic
- Cyanobacteria fix N₂ in HETEROCYSTS (oxygen-free)
- "Blue-green algae" = cyanobacteria = BACTERIA not algae
- Granules: S (purple S), carboxysome, gas vacuole = flotation

B. CHEMOAUTOTROPHS / CHEMOLITHOTROPHS (inorganic + CO₂)

- Nitrifying: NH₃ → NO₂⁻ (Nitrosomonas) → NO₃⁻ (Nitrobacter)
- Sulfur: H₂S → S → SO₄²⁻ (Thiobacillus, Beggiatoa)
- Iron: Fe²⁺ → Fe³⁺ (Ferrobacillus); Hydrogen: H₂ → H₂O
- All aerobes (need O₂) EXCEPT H₂-bact & methanogens (anaerobes)
- ID: "oxidizes a mineral for ATP, carbon = CO₂" = chemo-autotroph
- Nitrogen cycle: fixation (N₂ → NH₃) / nitrif (NH₃ → NO₃⁻) / denitrif (NO₃⁻ → N₂)
- N₂-fixers: Rhizobium (symbiont), Azotobacter, Clostridium, cyanobacteria
- Nitrogenase = O₂-sensitive → heterocyst protects it

C. CHEMOHETEROTROPHS (organic + organic C) = MOST COMMON

- 3 lifestyles: SAPROPHYTE (dead) / PARASITE (host) / SYMBIONT (mutualism)
- Examples: E. coli, Staphylococcus, Lactobacillus, Salmonella
- Glucose fates: aerobic / anaerobic respiration / fermentation
- ATP: aerobic ~30-38 / anaerobic moderate / fermentation 2
- Products: CO₂+H₂O / NO₂/H₂S/FeS / lactic + ethanol + CO₂
- ID: "must eat ready organic food" = chemo-heterotroph
- Decomposer = mineralizer = saprophyte (recycles C & N)
- Rhizobium = the ONE chemo-hetero that's also a N-fixer (symbiont)
- Synonyms: chemoorganotroph = chemoheterotroph = organotroph

D. O₂ REQUIREMENT (6 types) + mnemonic

- Obligate aerobe = NEEDS O₂ (Mycobacterium, Pseudomonas)
- Obligate anaerobe = O₂ KILLS (Clostridium, Methanogens)
- Facultative anaerobe = prefers O₂, can ferment (E. coli, Staphylo)
- Microaerophile = LITTLE O₂ only (Helicobacter, Campylobacter)
- Aerotolerant = no O₂ needed, not harmed (Lactobacillus, Strep)
- Photoaerotolerant = phototroph, O₂ OK (green/purple non-S)
- Mnemonic "O-OF-M-A": Obligate, Facultative, Micro, Aero (remember)
- "Obligate = forced; Facultative = can switch; Micro = small dose"
- ID: "O₂ kills it" = obligate anaerobe; "a little O₂" = microaerophile

CROSS-TYPE QUICK FACTS + EXAM GUTS

- 3 keys = ENERGY (light/chemo) - CARBON (CO₂/organic) - DONOR (inorganic/organic) = the 4 types
- "Photo + CO₂" = photoAUTOTroph (cyanobacteria); "Photo + organic C" = photoHETERO (Rhodospirillum)
- GOTCHA: cyanobacteria release O₂ (oxygenic); green/purple sulfur do NOT (anoxygenic, deposit S)
- "Autotroph" = makes own C from CO₂; "Heterotroph" = must eat organic C. "Litho" = inorganic donor; "organo" = organic
- Nitrogen flow: FIXATION (N₂&NH₃) → NITRIFICATION (NH₃&NO₃⁻) → DENITRIFICATION (NO₃⁻&N₂)
- GOTCHA: N₂ fixation needs O₂-FREE (nitrogenase) → Rhizobium in nodules, cyanobacteria in HETEROCYSTS
- Rhizobium is the ONE chemo-hetero that also FIXES N₂ (the exam "special" case)
- Methanogens are ARCHAEA (not bacteria) but fix: H₂ + CO₂ → CH₄ (anaerobes)

MNEMONICS + EXAM GUTS (re-identify every type in 30 s)

- "E-C-E" = Energy / Carbon / Electron-donor = the 3 letters naming a bacterium (photo/litho etc.)
- Types: PHL (photo-litho-auto) = cyanobacteria; CHO (chemo-organo-hetero) = most bugs; CTL (chemo-litho-auto) = nitrifiers/S
- O₂ "O-OF-M-A" = Obligate, Facultative, Microaerophile, Aerotolerant (+ photoaerotolerant)
- N-cycle "F-N-D" = FIXATION → NITRIFICATION → DENITRIFICATION (N₂&NH₃&NO₃⁻&N₂)
- GUT: "oxygenic vs anoxygenic" = split H₂O (O₂ out) vs H₂S (S out, no O₂) = THE photo-type exam
- GUT: "heterocyst" = cyanobacterial O₂-free cell for N₂ fixation; Rhizobium = nodules (legumes)
- Draw: Fig 1 (tree) + Fig 5 (nitrif/sulfur) + Fig 8 (glucose fates) + Fig 12 (heterocyst) = 4 exam figures
- 4-mark pattern: 4 types (energy/C/donor/examples); 4.8 = add one equation + one ecology role + labelled diagram

Fig 13: One-page outlet map of bacterial nutrition - every type, O₂, N-cycle, mnemonics and exam gotchas on one page.

Appendix - Similarity / Difference / Synonym bank (identify & explain)

Term	Synonyms	Identify-by
Photoautotroph	photolithoautotroph, phototroph	light + CO ₂ (cyanobacteria, S-bacteria)
Chemoautotroph	chemolithoautotroph, lithoautotroph	inorganic oxidation + CO ₂ (nitrifiers, S, Fe)

Photoheterotroph	photoorganotroph	light + organic C (purple/green non-S)
Chemoheterotroph	chemoorganotroph, organotroph	organic energy + C (most bugs, pathogens)
Saprophyte	decomposer, mineralizer	dead organic matter
Nitrification	ammonia -> nitrate	2 steps (Nitrosomonas -> Nitrobacter)
Denitrification	nitrate -> N ₂	anaerobic (Clostridium, Pseudomonas)
Heterocyst	N ₂ -fixing cell	cyanobacterial O ₂ -free differentiated cell
Obligate	forced / required	"-obligate" = must-have / must-avoid O ₂